

Mechanisms and Applications of $\gamma\delta$ T Cells in Anti-Tumor Immunity

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Abstract

$\gamma\delta$ T cells represent a distinctive subset of immune cells with considerable promise in cancer immunotherapy. They recognize a broad spectrum of tumor-associated antigens via non-major histocompatibility complex (non-MHC) pathways and exert antitumor effects by inducing apoptosis, directly lysing tumor cells, and modulating other immune components. This unique antigen-recognition capacity has spurred extensive efforts to harness $\gamma\delta$ T cells for innovative immunotherapeutic applications. Consequently, their use in cancer treatment is gaining increasing traction. Researchers have employed genetic engineering and other strategies to enhance $\gamma\delta$ T cell anti-tumor efficacy and have begun evaluating their potential in clinical trials. However, this therapeutic approach faces notable challenges, including interindividual variability in response and risk of adverse effects. Future research should aim to achieve a more comprehensive understanding of the mechanisms of $\gamma\delta$ T cells across different tumor types and improve their safety and efficacy in clinical settings. This review synthesizes recent advances in $\gamma\delta$ T cell research, examining their roles in tumor recognition, cytotoxicity, immunoregulation, and anti-tumor immunity. It further evaluates preclinical and clinical evidence to assess the therapeutic potential of $\gamma\delta$ T cell-based cancer immunotherapies.

Plain Language Summary

Our immune system includes a wide variety of cells that help defend the body against infections and diseases, including cancer. This article focuses on a special type of immune cell called $\gamma\delta$ T cells (pronounced “gamma-delta T cells”), which have unique abilities that make them promising tools for cancer treatment. Unlike other immune cells that need specific markers to recognize harmful cells, $\gamma\delta$ T cells can detect signs of stress or damage directly on tumor cells. This means they can respond faster and more broadly to different types of cancer. They also release proteins called cytokines that help coordinate the immune response and can directly kill cancer cells. Researchers are exploring different ways to use $\gamma\delta$ T cells to treat cancer. One approach involves collecting these cells from patients or healthy donors, growing them in the lab, and then putting them back into the patient’s body to attack tumors. Another method involves genetically engineering the cells to make them even better at recognizing and destroying cancer cells. Early studies show these treatments are safe, but scientists are still working on making them more effective. The article also discusses how the environment around a tumor—called the tumor microenvironment—can make it harder for immune cells to work. Researchers are now designing strategies to help $\gamma\delta$ T cells survive and stay active in these difficult conditions. While these therapies are still in the early stages of testing, they offer new hope for improving cancer treatment. With continued research, $\gamma\delta$ T cells could become a powerful part of future cancer therapies, especially when used together with other treatments like chemotherapy or immunotherapy.

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Keywords

$\gamma\delta$ T cells, tumor immunity, immunotherapy, anti-tumor, tumor microenvironment

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Introduction

According to the GLOBOCAN 2022 report, approximately 20 million new cancer cases were diagnosed globally, reflecting a steady rise compared to previous years and highlighting the persistent global burden of malignancies.^{1,2} Despite advances in surgery, radiotherapy, and chemotherapy, cancer continues to rank among the leading causes of mortality, primarily due to the complexity of tumor biology. Hallmarks such as sustained proliferative signaling, resistance to apoptosis, angiogenesis, and metastatic potential continue to constrain therapeutic efficacy.³ These obstacles have prompted the development of novel immunotherapeutic strategies aimed at overcoming the challenges posed by tumor heterogeneity and the immunosuppressive tumor microenvironment (TME).

$\gamma\delta$ T cells have garnered increasing interest due to their unique immunological characteristics. Unlike conventional $\alpha\beta$ T cells, $\gamma\delta$ T cells recognize stress-induced ligands and tumor-associated antigens independently of the major histocompatibility complex (MHC), allowing rapid immune responses without prior antigen processing. They exert multifaceted antitumor activities by producing cytokines, modulating the TME, and directly lysing tumor cells.^{4–8} These attributes make $\gamma\delta$ T cells an attractive candidate for cancer immunotherapy. However, their clinical translation has been modest compared to the notable successes of $\alpha\beta$ CAR-T cell therapy. Factors such as limited cell abundance, functional exhaustion, and immunosuppression within the TME remain significant obstacles to therapeutic efficacy.

Biological Characteristics of $\gamma\delta$ T Cells

Subpopulations and Identification of $\gamma\delta$ T Cells

$\gamma\delta$ T cells represent a distinct subset of T lymphocytes characterized by their γ and δ T-cell receptor (TCR) chains. Although they account for only 0.5%–5% of circulating T cells, they exhibit substantial heterogeneity in tissue distribution and functional specialization.⁹ Human $\gamma\delta$ T cells are broadly categorized into V δ 1, V δ 2, V δ 3, and V δ 5 subsets, each associated with different anatomical niches and modes of antigen recognition.¹⁰ (Table 1). V δ 1 T cells are predominantly found in tissues such as the intestinal epithelium and mucosal surfaces. They play a crucial role in maintaining the mucosal barrier and regulating immune responses, thereby contributing to overall immune homeostasis in these regions.¹¹ In contrast, V γ 9V δ 2 T cells predominate in the peripheral blood and uniquely detect metabolic stress in tumor cells

through phosphoantigen (pAg) recognition.^{12,13} V δ 3 T cells are primarily located in the small intestine and are associated with intestinal immune defense.¹⁴ V δ 1 and V δ 3 T cells recognize glycosphingolipids presented by MHC-related CD1 molecules (CD1b, CD1c, and CD1d) via their TCRs. Unlike V δ 1 T cells, V δ 3 T cells do not recognize CD1b and CD1c. These unconventional recognition pathways enable $\gamma\delta$ T cells to operate at the interface of innate and adaptive immunity, swiftly bridging antigen-independent sensing with antigen-specific responses.^{15,16} In contrast, V δ 5 T cells recognize the endothelial protein C receptor (EPCR), establishing a direct link to stress surveillance in epithelial malignancies.¹⁷ These subsets exhibit distinct functional roles and tissue distributions, thereby enhancing the complexity and adaptability of the immune system.

A central mechanism of V γ 9V δ 2 T cell activation involves the butyrophilin (BTN) family, particularly BTN3A1 and BTN2A1. Tumor cells with dysregulated mevalonate metabolism accumulate isopentenyl pyrophosphate (IPP) and related phosphoantigens (pAgs).^{18,19} Butyrophilin 3A1 (BTN3A1) is essential for the phosphoantigen (pAg)-mediated activation of V γ 9V δ 2 T cells. It functions as a sensor that detects intracellular pAgs and transmits activation signals to these T cells.^{20,21} Intracellular pAgs bind to the B30.2 domain of BTN3A1, inducing conformational changes that are relayed to the extracellular domain.^{22,23} BTN2A1 cooperates with BTN3A1 by directly engaging the V γ 9 chain, thereby stabilizing the immunological synapse between $\gamma\delta$ T cells and tumor targets.^{24,25} Carrie R. Willcox et al hypothesized that BTN3A1 detects pAgs by binding to the complementarity-determining region 3 (CDR3) of the TCR via a surface patch on its IgV domain, thereby facilitating the formation of a stable complex with BTN2A1. This pAg-induced surface V γ 9V δ 2 TCR complex is critical for effective phosphoantigen (pAg) recognition.²⁶ The recognition and response of V γ 9V δ 2 T cells to pAgs through the BTN3A1–BTN2A1 complex underscore their specialized role in immune surveillance (Figure 1). Their unique mechanism of detecting dysregulated tumor metabolism makes them promising targets for cancer immunotherapy. However, the clinical translation of these distinctive recognition pathways remains limited. Although V γ 9V δ 2 cells are capable of detecting phosphoantigen accumulation in tumor cells, their frequency declines with age and chronic stimulation, thereby reducing their therapeutic potential.²⁷ Similarly, although V δ 1 cells exhibit promising tissue-resident tumor reactivity, standardized expansion protocols remain less well established compared to those of V γ 9V δ 2 cells. Thus, dissecting recognition mechanisms such as the BTN3A1–BTN2A1 axis is

Table 1. $\gamma\delta$ T Cell Subsets and Tumor Immunity.

δ chain	γ chain	Distribution	Tumor immune function
V δ 1	V γ 2, V γ 3, V γ 4, V γ 5, V γ 8 and V γ 9	Peripheral blood, gut, and other epithelial tissues	Recognize tumor antigens, immune surveillance, directly kill tumor cells, immunomodulation
V δ 2	V γ 9	Peripheral blood and liver	Recognize tumor antigens, immune surveillance, directly kill tumor cells, immunomodulation
V δ 3	V γ 2 and V γ 3	Peripheral blood, liver, and gut	Recognize tumor antigens, directly kill tumor cells, immunomodulation
V δ 5	V γ 4	Peripheral blood	Recognize tumor antigens

essential not only for elucidating $\gamma\delta$ T cell biology but also for optimizing their therapeutic application.

Anti-tumor Ability of $\gamma\delta$ T Cells

$\gamma\delta$ T Cells can Affect the Production of Multiple Immune Cytokines

The tumor microenvironment (TME) comprises diverse cell populations, soluble signaling molecules, blood vessels, and extracellular matrix components, all of which influence tumor progression and therapeutic outcomes. Research has demonstrated that $\gamma\delta$ T cells regulate the TME and exert antitumor effects via multiple mechanisms. These cells have been identified in the tumor microenvironment of melanoma, breast, colon, lung, and ovarian cancers.²⁸

One of the primary functions of $\gamma\delta$ T cells is the regulation of cytokine production. Cytokines are small proteins that serve as intracellular communication signals within the immune system. These molecules are fundamental to immune regulation, mediating both inflammatory promotion and suppression. By modulating cytokine levels, $\gamma\delta$ T cells can influence immune responses and potentially control tumor growth. $\gamma\delta$ T cells produce multiple cytokines, including IL-17, IFN- γ , IL-4, IL-10, and TGF- β .^{29–32}

The IL-17 signaling pathway is initiated through the SEFIR domain following TCR activation, which subsequently triggers MAPK, NF- κ B, and C/EBP pathways to regulate IL-17 production by $\gamma\delta$ T-17 cell.³³ ROR γ t serves as the master transcription factor for IL-17A, controlling its expression not only at the transcriptional level but also via epigenetic modifications, suggesting potential druggable targets for future therapies.³⁴ Additional signaling cascades remain incompletely elucidated, underscoring the complexity of IL-17 regulation in $\gamma\delta$ T cells. Functionally, IL-17A is robustly secreted by $\gamma\delta$ T-17 cells and has been implicated in the pathogenesis of several cancers, including colorectal, hepatocellular, lung, and nasopharyngeal carcinomas. Mechanistically, IL-17A can (a) recruit neutrophils and myeloid-derived suppressor cells (MDSCs), (b) promote angiogenesis, and (c) upregulate tumor PD-L1 and survival pathways, thereby facilitating tumor progression and therapy resistance.^{7,35–39} In

nasopharyngeal carcinoma, for instance, tumor-derived exosomes can induce $\gamma\delta$ T cell differentiation into $\gamma\delta$ T-17 cells, which in turn enhance radioresistance and diminish the efficacy of radiotherapy.⁴⁰ Conversely, IL-17 can exert antitumor effects in specific contexts by promoting CD8⁺ T-cell infiltration and synergizing with pro-inflammatory cues to eliminate early-stage transformed cells or pathogens.⁷ Its biological consequences are highly context-dependent: evidence across tumor types demonstrates both antitumor and protumor activities, highlighting the importance of considering cytokine source, temporal dynamics, and tumor-intrinsic signaling when designing $\gamma\delta$ T cell therapies.

IFN- γ is a canonical antitumor cytokine: it enhances antigen presentation, promotes M1 polarization, increases tumor immunogenicity, and augments CD8⁺ and NK cell functions.⁴¹ In $\gamma\delta$ cells, IFN- γ production is controlled by T-bet/Eomes balance and can be induced via STING-TBK1 signaling. T-bet promotes the differentiation of $\gamma\delta$ T cells into IFN- γ -producing cells, whereas Eomes inhibits IFN- γ production. The dynamic interplay between T-bet and Eomes determines the differentiation trajectory of $\gamma\delta$ T cells.^{42,43} This transcriptional balance contributes to antitumor activity via multiple mechanisms, including tumor cell apoptosis induction, proliferation suppression, and differentiation promotion. Additionally, IFN- γ also remodels the tumor microenvironment by attracting effector immune cells, regulating angiogenesis, and enhancing antigen presentation through upregulation of MHC molecules. These effects collectively improve tumor immunogenicity and can increase responsiveness to chemotherapy and radiotherapy by facilitating drug delivery and sensitizing cancer cells to cytotoxic damage.^{44,45} However, the antitumor effects of IFN- γ are counterbalanced by its potential to promote immune evasion. Chronic or sustained IFN- γ exposure can paradoxically induce adaptive resistance, enabling tumor cells to evade immune surveillance.⁴¹ Mechanistically, tumors may upregulate PD-L1, IDO, and other inhibitory pathways, or alter antigen-processing machinery to escape CTLs and NK cells.^{44–46} Thus, IFN- γ not only facilitates tumor rejection but also promotes the selection of resistant clones and fosters immunosuppressive microenvironments. Given this duality, therapeutic strategies involving IFN- γ require careful calibration.

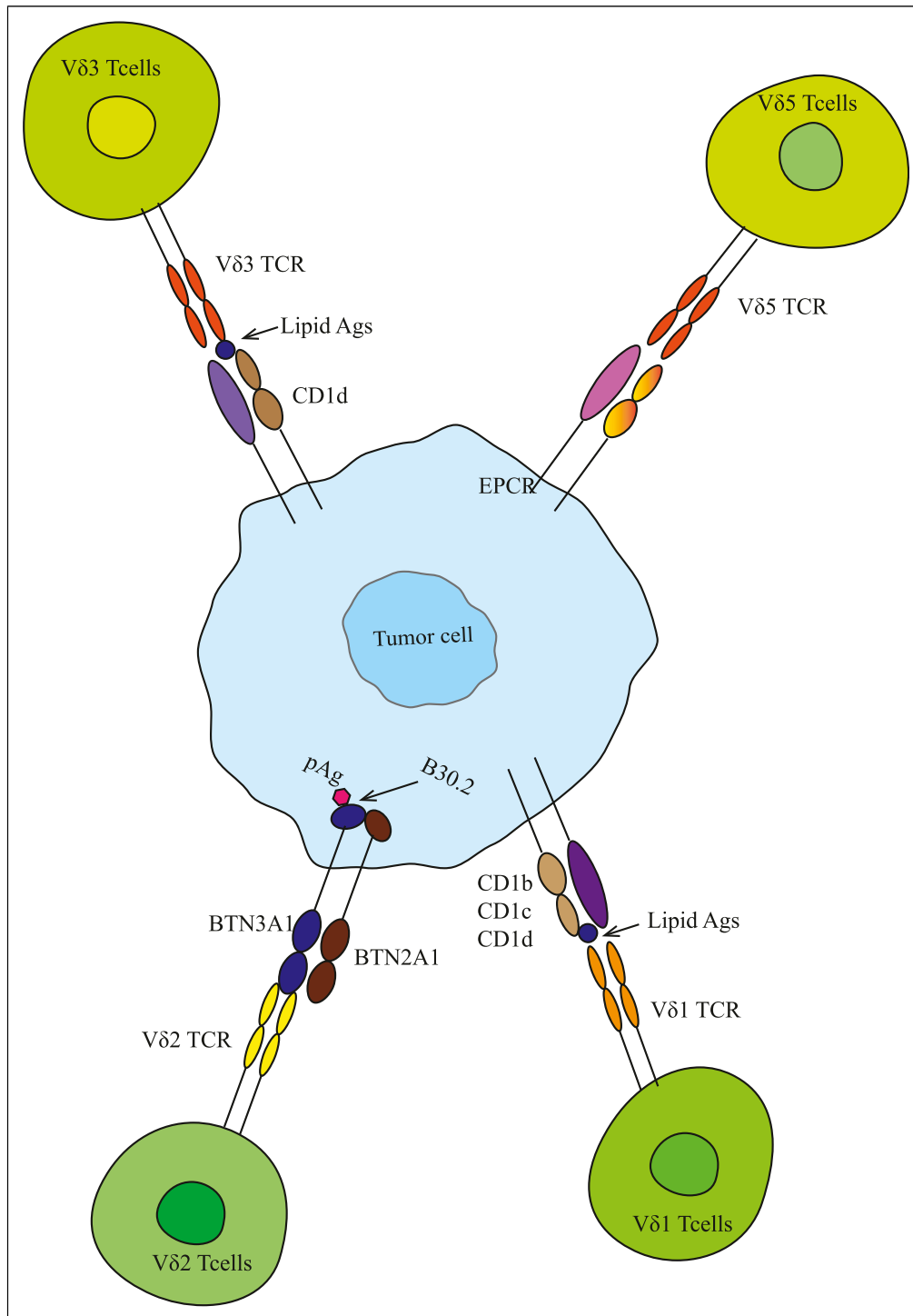


Figure 1. Ligands Recognized by Human $\gamma\delta$ T Cells. V γ 9 V δ 2 T Cells Sense Phosphoantigens (pAgs) Derived From the Mevalonate Pathway via BTN3A1 and BTN2A1. V δ 1 T Cells Recognize Lipid Antigens Presented by CD1 Molecules (CD1b, CD1c, CD1d), V δ 3 T Cells Also Engage CD1d. V δ 5 T Cells Detect Endothelial Protein C Receptor (EPCR) Expressed on Stressed or Transformed Cells

Approaches that generate acute, localized IFN- γ bursts—for example, $\gamma\delta$ T cells engineered to secrete IFN- γ upon antigen recognition—may harness its protective effects while limiting chronic toxicity.⁴⁷ Moreover, combining IFN- γ -enhancing

therapies with immune checkpoint inhibitors (eg, PD-1/PD-L1 or IDO blockade) represents a rational strategy to counteract IFN- γ -induced adaptive resistance. Biomarkers of IFN- γ pathway activation may further inform patient selection,

optimizing therapeutic efficacy while minimizing associated risks.

IL-4 is a pleiotropic cytokine secreted by activated T cells, basophils, and mast cells, and is primarily associated with Th2-type immune responses.⁴⁸ IL-4 signaling activates two major pathways: the STAT6 transcription factor pathway and the insulin receptor substrate (IRS)-1/2 pathway.⁴⁹ In $\gamma\delta$ T cells, IL-4 suppresses activation via a STAT6-dependent mechanism, thereby restraining their cytotoxic potential.⁵⁰ Beyond T cell regulation, IL-4 and its receptor influence apoptosis, chemotherapy sensitivity, and overall cancer prognosis.⁵¹ Within the tumor microenvironment, IL-4 predominantly exerts protumorigenic effects. It drives the polarization of tumor-associated macrophages toward an M2-like phenotype, which suppresses effective antitumor immunity and fosters tumor progression.⁵² For example, in colorectal cancer cell lines (HT-29 and DLD-1), IL-4 upregulates NADPH oxidase 1 activity, promoting cell growth; in hepatocellular carcinoma, IL-4 contributes to tumor progression; and in gastric cancer cells, IL-4 signaling can modulate cell-cycle progression.^{53,54} However, IL-4's role is not uniformly protumorigenic: in specific contexts, such as in CRL1739 gastric cancer cells, IL-4 binding to its receptor induces G1 cell cycle arrest, indicating a context-dependent growth-inhibitory function.⁵⁵ This duality highlights IL-4 as both a mediator of tumor-supportive immune suppression and, in rare cases, a regulator of tumor cell proliferation with inhibitory potential. From a translational perspective, IL-4/IL-4R blockade or inhibition of downstream STAT6 signaling has been proposed as a strategy to counteract its tumor-promoting functions. Conversely, elucidating the conditions under which IL-4 induces antiproliferative responses may uncover novel therapeutic opportunities. Thus, IL-4 exemplifies the complex, context-dependent cytokine networks that must be precisely dissected to optimize both $\gamma\delta$ T cell-based and broader immunotherapeutic strategies.

In addition to pro- and anti-tumor cytokines such as IL-17, IFN- γ , and IL-4, immunoregulatory cytokines further influence $\gamma\delta$ T cell function and cancer progression. V δ 1 T cells, for example, produce high levels of IL-10, which potently suppresses the cytotoxicity of V γ 9V δ 2 T cells and thereby constrains their antitumor efficacy.⁵⁶ TGF- β is another key immunoregulatory factor secreted by V δ 1 cells, with broad implications for tumor biology. While TGF- β regulates $\gamma\delta$ T cell development, it also promotes tumor progression by inducing epithelial-mesenchymal transition (EMT) in epithelial cells, thereby enhancing invasion and metastasis.^{42,57,58} Beyond tumor progression, TGF- β plays a central role in shaping tissue-resident $\gamma\delta$ T cell populations. In the gut, for instance, TGF- β regulates the development of TCR $\gamma\delta^+$ CD8 $\alpha\alpha^+$ intraepithelial lymphocytes (IELs) through Smad2- and Smad3-dependent pathways, which are essential for maintaining mucosal homeostasis.⁵⁹ From a therapeutic perspective, selectively inhibiting suppressive cytokine signaling or engineering $\gamma\delta$ T cells to resist IL-10 and TGF- β may help

preserve cytotoxic function while maintaining mucosal integrity.

Cytotoxic Effects on Malignant Tumors

The cytotoxic machinery of $\gamma\delta$ T cells induces tumor cell apoptosis through multiple, non-MHC-restricted pathways. V δ 1 $\gamma\delta$ T cells eliminate tumor and lymphoma cells by recognizing stress-induced ligands such as MICA, MICB, and UL16-binding proteins (ULBPs) expressed on malignant cells.⁶⁰ V γ 9 V δ 2 T cells exhibit potent cytotoxic activity against a broad range of malignancies via at least two major pathways: (i) TCR-mediated recognition of endogenous phosphoantigens such as isopentenyl pyrophosphate (IPP), as well as ectopically expressed mitochondrial F1-ATPase or the F1-ATPase/ApoA1 complex on tumor membranes, which stabilize TCR binding^{19,61,62}; and (ii) NKG2D engagement with MICA, MICB, or ULBPs, which activates a DAP10-dependent signaling cascade that triggers the release of perforin, granzymes, and other cytotoxic mediators (Figure 2).^{63–67} Importantly, NKG2D signaling can override inhibitory NK receptor pathways, enabling $\gamma\delta$ T cells to retain cytotoxic function even under immunosuppressive conditions. Despite the robustness of these cytotoxic mechanisms, their *in vivo* efficacy is frequently attenuated. Hypoxia and nutrient deprivation within the tumor microenvironment (TME) impair degranulation, many tumors downregulate Fas or shed soluble MICA/B to evade NKG2D recognition, and immunosuppressive cytokines such as TGF- β limit $\gamma\delta$ T-cell effector function. Moreover, many of these cytotoxic pathways are shared with NK and $\alpha\beta$ T cells, raising questions about whether $\gamma\delta$ T cells confer unique advantages in clinical applications. To fully harness their therapeutic potential, translational strategies should include blocking ligand shedding, enhancing cell persistence, and engineering $\gamma\delta$ T cells to resist tumor microenvironment-derived inhibitory signals. Such approaches may translate their inherent cytotoxic potential into meaningful clinical benefit.

Treatment Strategies

Autologous $\gamma\delta$ T Cell Adoptive Transfer Therapy

Adoptive transfer therapy using autologous $\gamma\delta$ T cells has shown considerable promise in cancer immunotherapy by leveraging their capacity to directly lyse tumor cells, secrete antitumor cytokines, and activate other immune subsets. Both V δ 1 and V δ 2 subsets have been investigated for clinical application, supported by advances in *ex vivo* expansion technologies.⁶⁸ For V δ 1 cells, artificial antigen-presenting cells expressing CD40 L and the cytomegalovirus antigen pp65 have been used as feeder layers to promote proliferation and enhance cytotoxicity,⁶⁹ while newer protocols combining anti-CD3 (OKT-3) with IL-15 allow rapid expansion of highly pure V δ 1 populations amenable to CAR engineering.⁷⁰ Notably, Delta One T (DOT) cells—expanded V δ 1 cells enriched for NK

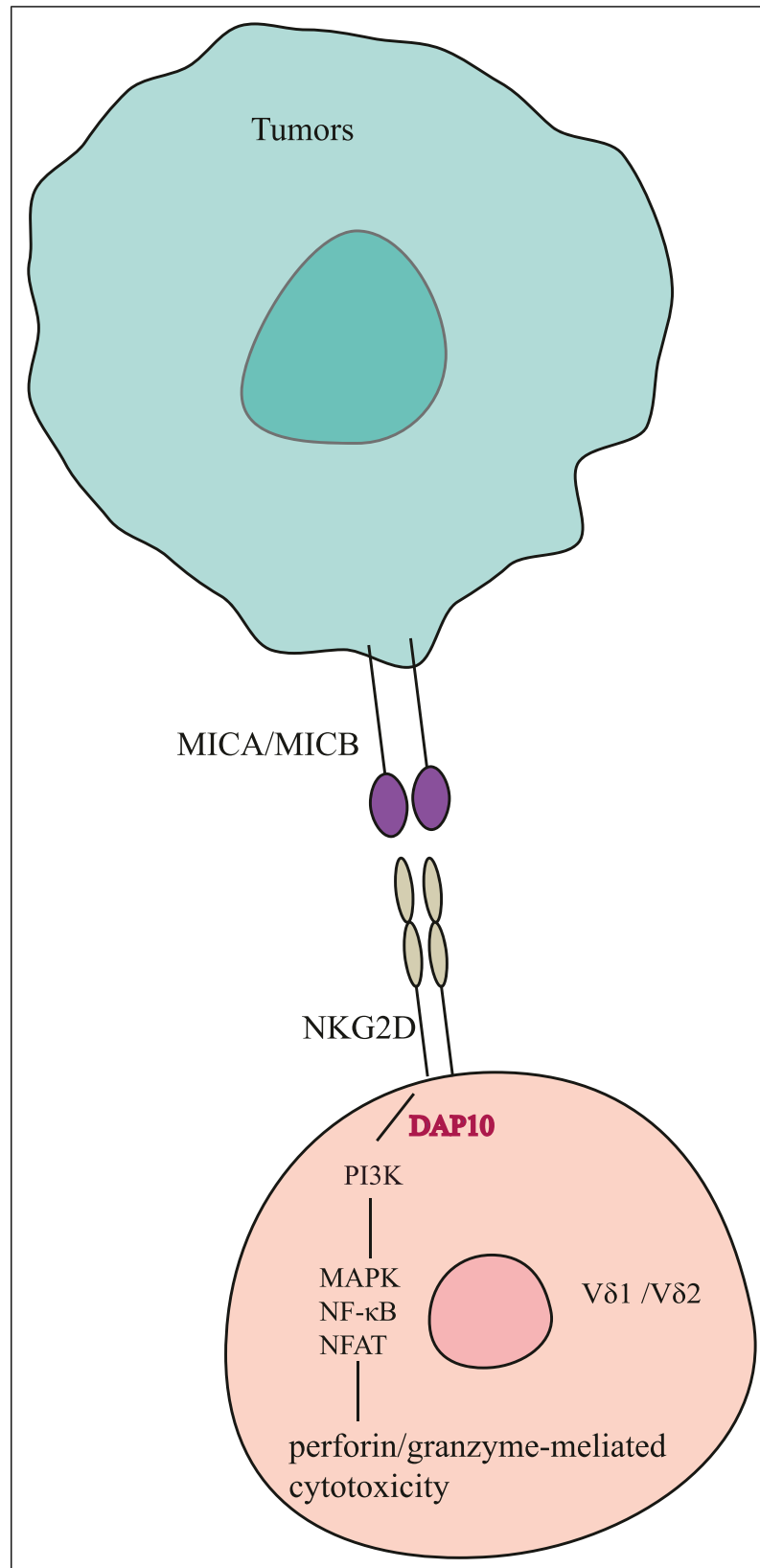


Figure 2. Engagement of NKG2D With MICA/MICB Triggers DAP10-dependent Recruitment of PI3K and Grb2/Vav1, Leading to Downstream Activation of MAPK, NF-κB, and NFAT Signaling Cascades. This Results Cytokine Secretion (IFN-γ, TNF-α, IL-17), and Perforin/Granzyme-Mediated Cytotoxicity

receptors—have shown potent activity against colorectal and other solid tumors.⁷¹ By contrast, V δ 2 cell expansion primarily relies on phosphoantigen stimulation. Aminobisphosphonates such as pamidronate or zoledronic acid (ZOL) induce intracellular accumulation of isopentenyl pyrophosphate, thereby selectively activating and expanding V γ 9V δ 2 T cells.^{18,19}

Clinical studies have established proof of feasibility. Wilhelm et al⁷² reported that pamidronate combined with low-dose IL-2 yielded limited expansion, whereas sustained high-dose IL-2 administration achieved partial remissions or stable disease in some patients. Dieli et al⁷³ demonstrated that ZOL plus IL-2 was superior to ZOL alone in metastatic prostate cancer. More recently, Bacillus Calmette–Guérin (BCG) was shown to synergize with ZOL in enhancing V δ 2 expansion and polyfunctionality, offering a potential alternative approach.^{74,75} These trials demonstrated that autologous $\gamma\delta$ T cells could be safely administered and were capable of exerting measurable antitumor activity, with some patients showing partial responses or disease stabilization. Importantly, $\gamma\delta$ T cells possess a favorable safety profile compared with $\alpha\beta$ T cells, as they rarely cause graft-versus-host disease (GVHD).^{76s}

However, clinical outcomes of autologous $\gamma\delta$ T cell therapy have been modest, and durable responses remain rare. A key limitation is the low frequency of $\gamma\delta$ T cells in peripheral blood, which restricts ex vivo expansion. Even when expanded, transferred $\gamma\delta$ T cells often display limited persistence in vivo, partly due to exhaustion and suppression by the tumor microenvironment (TME). Tumors may downregulate stress ligands, secrete inhibitory cytokines such as TGF- β , and induce metabolic constraints that further impair $\gamma\delta$ T-cell cytotoxicity. In addition, repeated stimulation with aminobisphosphonates may induce functional anergy in V γ 9V δ 2 cells, further limiting therapeutic efficacy.⁷⁷

To enhance the clinical potential of autologous $\gamma\delta$ T-cell adoptive transfer, several strategies are under investigation. These include optimizing ex vivo culture conditions to preserve effector function, combining $\gamma\delta$ T cell infusion with immune checkpoint blockade or cytokine support, and engineering $\gamma\delta$ T cells with CAR constructs or resistance to TME-derived inhibitory signals.⁷⁸ Collectively, while autologous $\gamma\delta$ T cell therapy has demonstrated safety and feasibility, significant translational challenges remain, and its future success will hinge on overcoming barriers to cell persistence, functional fitness, and resistance to tumor-mediated suppression.

Allogeneic $\gamma\delta$ T Cell Adoptive Transfer Therapy

Allogeneic $\gamma\delta$ T cell adoptive transfer therapy, in which $\gamma\delta$ T cells are harvested from healthy donors and expanded for infusion into patients, has emerged as an appealing alternative to autologous approaches. This strategy builds on the favorable safety profile of $\gamma\delta$ T cells, which seldom provoke graft-versus-host disease, thereby enabling the development of standardized “off-the-shelf” cell products.^{79,80} Early clinical studies have demonstrated both feasibility and safety. For example, Kondo et al⁸¹ successfully expanded $\gamma\delta$ T cells from six healthy donors

and confirmed their significant antitumor activity in patients, underscoring the translational potential of this approach. The combination of zoledronic acid (ZOL) and low-dose interleukin-2 (IL-2) remains a standard method for V γ 9V δ 2 T cell expansion: ZOL promotes accumulation of endogenous phosphoantigens in tumor cells, while IL-2 boosts $\gamma\delta$ T cell proliferation and cytokine secretion.⁸² Compared with autologous transfer, allogeneic $\gamma\delta$ T cell therapy is associated with fewer adverse effects and greater scalability. Despite these advantages, several obstacles limit the success of allogeneic $\gamma\delta$ T-cell therapy. Persistence of infused cells in vivo remains suboptimal, reducing the likelihood of durable responses. Tumor immune evasion mechanisms—including shedding of NKG2D ligands (MICA/B), secretion of immunosuppressive cytokines such as TGF- β , and induction of metabolic stress—further impair their effector function. Moreover, while $\gamma\delta$ T cells are generally well tolerated, the long-term safety of large-scale allogeneic products remains uncertain, with potential risks of off-target cytotoxicity and immune dysregulation necessitating careful monitoring.

To address these challenges, innovative engineering strategies are under active exploration. These include CAR-modified $\gamma\delta$ T cells to enhance tumor recognition, IL-15 armoring to boost survival and proliferation, and genetic modification to resist tumor microenvironment-derived suppression.⁸³ Advances in biomanufacturing are also enabling large-scale, GMP-compliant expansion of $\gamma\delta$ T cells, paving the way for broader clinical testing. Compared with autologous $\gamma\delta$ T-cell transfer, the allogeneic approach offers clear logistical and scalability advantages, but its long-term efficacy and safety still require validation in well-designed clinical trials.⁸⁰ Ultimately, the future of allogeneic $\gamma\delta$ T cell therapy will hinge on balancing its “off-the-shelf” practicality with strategies that ensure persistence, potency, and durable tumor control.

Genetic Engineering Therapy

Genetic engineering strategies represent a critical frontier in the $\gamma\delta$ T cell–based cancer immunotherapy, with the goal of surmounting the limitations of conventional adoptive transfer. Among these, chimeric antigen receptor (CAR)-engineered $\gamma\delta$ T cells have garnered increasing attention. By combining the innate, non-MHC-restricted recognition capacity of $\gamma\delta$ T cells with CAR-mediated antigen specificity, CAR- $\gamma\delta$ T cells can target both stress ligands and tumor-associated antigens simultaneously.⁷⁸ In preclinical models, CAR-engineered V δ 1 and V γ 9V δ 2 cells have exhibited enhanced cytotoxicity against hematologic and solid tumors while retaining a favorable safety profile owing to the low risk of graft-versus-host disease.⁸⁴ Currently, CAR-T therapy has evolved through six generations, yet second-generation CAR-T cells remain dominant because of their proven stability and well-validated efficacy. A key advantage of $\gamma\delta$ T cells is their expression of the NKG2D receptor, which functions independently of major histocompatibility complex (MHC) molecules. This MHC independence enables CAR- $\gamma\delta$ T cells to maintain cytotoxic

Table 2. Preclinical Advances in CAR- $\gamma\delta$ T Cell Therapies.

Combination therapy	Disease	References
mRNA electroporation technology to transform $\gamma\delta$ T cells	Cancer	105
Treatment combining chemotherapy with cellular immunotherapy	Gastric cancer, esophageal cancer	106,107
Treatment of V γ 9V δ 2 T cells with IDO1 inhibitor	Triple-negative breast cancer (TNBC)	108
V δ 2 T cells combined with inhibiting B7-H3	Colon cancer	109
$\gamma\delta$ -T-Exos synergized with radiotherapy	Nasopharyngeal carcinoma	99
$\gamma\delta$ T cells as synergistic therapeutic cells for CD3-BsAbs	Pancreatic cancer	110
combined $\gamma\delta$ -T exosomes and photodynamic therapy (PDT)	Melanoma	111
Chitosan nanoparticles (CSNPs) were used to enhance the anti-tumor immunity of $\gamma\delta$ T cells	Cancer	112

activity against tumor cells, as ligands for NKG2D are frequently upregulated on malignant cells.⁸⁵ For example, CD19 is a commonly expressed antigen in B-cell malignancies, and anti-CD19 CAR-T cells have been specifically designed to target and eliminate these malignant cells.^{86,87} Treating solid tumors poses formidable challenges, as the tumor microenvironment (TME) obstructs CAR-T cell trafficking to tumor sites and hampers their capacity to mount effective immune responses. The TME further complicates targeting by erecting both physical and immunosuppressive barriers that impede engineered T cell infiltration and function. To overcome these obstacles, researchers have developed strategies to genetically modify T cell chemotaxis and tissue homing capabilities, as well as to engineer CAR-T cells that can directly target components of the tumor stroma.⁸⁸ Concurrently, to counteract the enhanced glycolytic activity and impaired metabolism observed in CAR-T cells, scientists have improved CAR-T cell persistence by reprogramming T cell metabolism through advanced CAR design.⁸⁹ Outlined in Table 2 below.

Nevertheless, substantial challenges must be addressed before $\gamma\delta$ T cell genetic engineering can transition to broad clinical application. Manufacturing protocols are inherently more complex than those for $\alpha\beta$ CAR-T cells, due to subset heterogeneity and inconsistent expansion success. Moreover, although early-phase studies indicate encouraging safety, large-scale clinical validation remains limited, and the durability of engineered $\gamma\delta$ T cell responses in solid tumors is still uncertain. Another critical concern is the functional overlap with NK and $\alpha\beta$ T cells, which raises doubts about whether $\gamma\delta$ T cell engineering truly delivers unique therapeutic advantages.⁹⁰ Future investigations should refine engineering platforms to maximize persistence, resist TME suppression, and harness the unique biological characteristics of $\gamma\delta$ T cells. Promising strategies include integrating CAR- $\gamma\delta$ T cells with immune checkpoint blockade, combining with oncolytic viruses or bispecific antibodies, and developing standardized allogeneic products. Collectively, genetic engineering offers the most compelling path toward unlocking the full

CAR- $\gamma\delta$ T subtype	Target antigen	Cancer type(s)	Preclinical outcome	References
V δ 1 CAR- $\gamma\delta$ T cells	CD19	B-cell acute lymphoblastic leukemia (B-ALL)	Robust cytotoxicity in vitro and in vivo, effective tumor clearance in xenograft models, and no GVHD observed	91
CAR-engineered human $\gamma\delta$ T cells	Retroviral CAR modification of $\gamma\delta$ T cells; retained cross-presentation ability	Broad tumor targets	Demonstrated enhanced cytotoxicity against tumor cells, maintained cross-presentation capacity, supporting $\gamma\delta$ T cells as versatile CAR platforms	92
PSCA-CAR-engineered V δ 1 $\gamma\delta$ T cells	PSCA	Pancreatic cancer	Demonstrated strong cytotoxicity and tumor control with a favorable safety profile compared to $\alpha\beta$ CAR-T; preclinical data suggest translational potential	93
Tandem CD33-CLL1 CAR-T	CD33 + CLL1	Acute myeloid leukemia (AML)	Demonstrated improved efficacy over single-target CARs, reduced relapse, and enhanced durability; provides insights for future $\gamma\delta$ CAR-T multi-target strategies	94
mRNA-engineered CD5-CAR- $\gamma\delta$ T (CD5 ⁺)	CD5	T-cell acute lymphoblastic leukemia (T-ALL)	mRNA-engineered CD5 ⁺ $\gamma\delta$ T cells overcame fratricide, showed strong cytotoxicity against T-ALL in vitro and in vivo	95

therapeutic potential of $\gamma\delta$ T cells, but its success will depend on translating preclinical advances into reproducible and durable clinical benefit.

Other Therapies

Small extracellular vesicles (EVs) have emerged as important mediators of intercellular communication, transferring proteins, nucleic acids, and metabolites that influence tumor progression and therapeutic responses.^{96,97} Their natural biocompatibility and stability make them attractive candidates for cancer vaccine development. Notably, $\gamma\delta$ T cell–derived EVs ($\gamma\delta$ T-EVs) appear to preserve dual antitumor functions of their parental cells by carrying cytotoxic molecules (eg, perforin, granzyme B) and immunostimulatory signals.⁹⁸

Recent studies provide proof of concept: Wang and colleagues reported that $\gamma\delta$ T-EVs enriched in interferon- γ (IFN- γ) and CD40 ligand (CD40 L) could transfer tumor-associated antigens (TAAs) to dendritic cells (DCs), thereby promoting antigen presentation and amplifying adaptive immune responses. Moreover, allogeneic $\gamma\delta$ T-EV vaccines derived from healthy donors exhibited antitumor efficacy comparable to autologous sources, suggesting a scalable and lower-complexity manufacturing platform.⁹⁹

Combination Therapy

PD-1 (programmed cell death protein 1, CD279) is a key immune checkpoint receptor expressed on T cells, B cells, NK cells, and myeloid populations. By binding to its ligands, PD-L1 or PD-L2, PD-1 suppresses T cell activation and proliferation, maintaining immune tolerance under physiological conditions.¹⁰⁰ Tumors exploit this pathway by upregulating PD-L1, enabling evasion of immune surveillance and attenuation of antitumor immunity.¹⁰¹ Importantly, the inhibitory PD-1/PD-L1 axis within the tumor microenvironment also constrains $\gamma\delta$ T cell activity, thereby limiting their therapeutic potential in vitro.¹⁰¹ Combining $\gamma\delta$ T cells with PD-1/PD-L1 blockade has shown promise: in an immunodeficient ovarian cancer model, $\gamma\delta$ T cells engineered to secrete anti-PD-1 antibodies—so-called “armored” $\gamma\delta$ T cells—exhibited enhanced cytotoxicity and favorable safety in vivo.¹⁰² Beyond PD-1, other checkpoint targets are relevant; for example, HLA-G is highly expressed in many solid tumors, and dual blockade of HLA-G and PD-L1 has been proposed to overcome TME-mediated resistance and achieve superior antitumor efficacy.¹⁰³ These findings strengthen the rationale for integrating $\gamma\delta$ T cell therapy with immune checkpoint inhibition. Furthermore, $\gamma\delta$ T cells may synergize with conventional modalities—including antibody therapy, chemotherapy, and radiotherapy—thus enhancing overall clinical benefit (Table 2).¹⁰⁴

Future Directions

Despite compelling preclinical data, $\gamma\delta$ T cell–based immunotherapies remain in an early stage of clinical translation and

lag behind $\alpha\beta$ CAR-T therapies in both efficacy and clinical uptake. To realize their full therapeutic potential, several critical challenges must be surmounted. First, the low abundance and heterogeneity of $\gamma\delta$ T cells complicate their large-scale expansion and standardization. While protocols for V γ 9V δ 2 expansion using aminobisphosphonates and IL-2 are well established, they often lead to peripheral depletion and functional exhaustion. More robust methods—such as V δ 1 cell expansion using artificial antigen-presenting cells, cytokine engineering, and feeder-free bioreactors—are urgently needed to ensure consistency and scalability. Second, the tumor microenvironment (TME) persists as a formidable barrier. Immunosuppressive cytokines (eg, TGF- β , IL-10), metabolic stress, and inhibitory checkpoint molecules (PD-1/PD-L1, HLA-G) can severely dampen $\gamma\delta$ T cell cytotoxicity. To counteract these mechanisms, strategies such as armored $\gamma\delta$ T cells engineered to express checkpoint inhibitors, dual-target CAR constructs (for example, PSCA + PD-L1), and cytokine tethering (eg, IL-15, IL-21) are being explored to bolster persistence without incurring systemic toxicity. Third, the dual roles of cytokines (such as IL-17, IFN- γ , and IL-4) underscore the need for subtype-specific modulation. Selective expansion of cytotoxic subsets (V δ 2 effector cells) while suppressing pro-tumor subsets ($\gamma\delta$ T-17) will be essential to avoid paradoxical effects like therapy resistance. This calls for biomarker-driven patient stratification and the development of $\gamma\delta$ T cell–specific immunophenotyping tools. Finally, integrating $\gamma\delta$ T cells into multimodal regimens—such as checkpoint inhibition, radiotherapy, oncolytic viruses, and exosome-based therapies—holds substantial promise. However, combination approaches must be rationally designed, as certain interactions (eg, IL-17–mediated radioresistance) can undermine efficacy. Preclinical models that better recapitulate the human TME, along with early-phase clinical trials incorporating correlative biomarker studies, will be crucial to define optimal combinations. In conclusion, the future of $\gamma\delta$ T cell therapy hinges not only on technological innovations in expansion and engineering but also on achieving deeper insight into their intricate biology within the TME. Overcoming these challenges via precise subset targeting, rational combination strategies, and biomarker-guided clinical development will be pivotal to elevate $\gamma\delta$ T cells from experimental modalities to reliable, scalable pillars of cancer immunotherapy.

Conclusion

In summary, $\gamma\delta$ T cells hold significant promise in cancer treatment. Due to their unique antigen recognition capabilities and multifunctional effector functions, they play a crucial role in anti-tumor immunity. $\gamma\delta$ T cells differentiate into diverse subpopulations, each exhibiting distinct distributions across tumor types. These subsets secrete characteristic cytokines with either antitumor or protumor effects and actively modulate the tumor microenvironment. Despite growing interest,

our understanding of $\gamma\delta$ T cell subset functionality remains incomplete, and the complexity of their roles across tumor types warrants further investigation. Compared to conventional $\alpha\beta$ T cells, $\gamma\delta$ T cells confer a lower risk of graft-versus-host disease (GVHD) in allogeneic settings, enhancing their appeal in cancer immunotherapy. However, $\gamma\delta$ T cell adoptive cell therapy (ACT) still faces challenges such as low cell expansion efficiency, poor in vivo persistence, insufficient tumor infiltration, and suppression by the immunosuppressive tumor microenvironment. The introduction of chimeric antigen receptor (CAR) technology has expanded the therapeutic potential of $\gamma\delta$ T cells by enhancing their targeting specificity and sustaining antitumor activity, though it has not fundamentally resolved the aforementioned limitations. Single-agent cellular therapies frequently fail to overcome the complexity and dynamism of the tumor microenvironment. Consequently, combination strategies are emerging as a critical avenue to augment $\gamma\delta$ T cell efficacy. Through synergistic effects with immune checkpoint inhibitors, chemotherapy, radiotherapy, or other immune cells, the proliferation, tumor infiltration, and functional activity of $\gamma\delta$ T cells can be significantly improved. With ongoing optimization and clinical validation of these combined strategies, $\gamma\delta$ T cells are expected to play a more prominent role as core immune effector cells within multimodal anti-tumor treatment regimens, advancing the development of precision tumor immunotherapy.

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Ethical Considerations

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